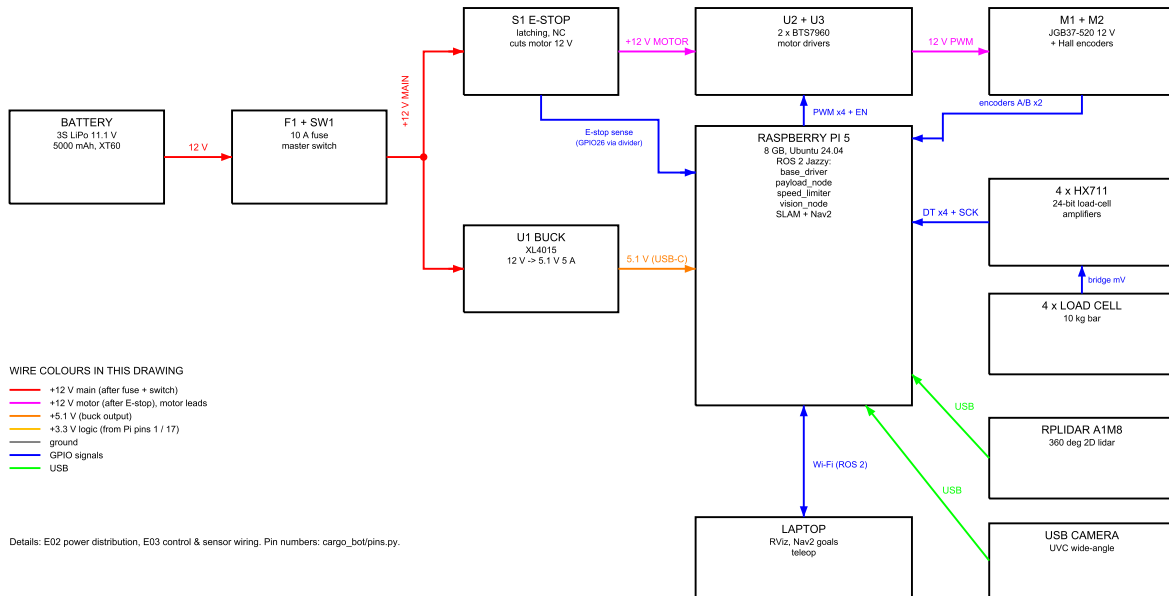
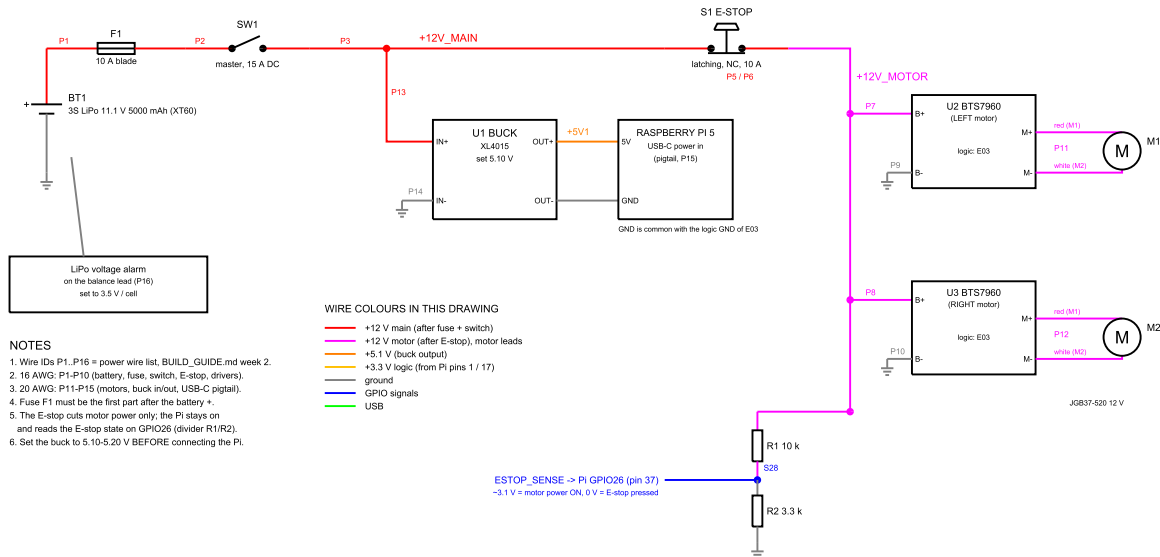


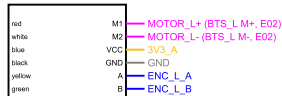


PROJECT						
AUTONOMOUS CARGO ROBOT - DESIGN v2						
TITLE				DWG No.		
SYSTEM BLOCK DIAGRAM				CR-V2-E01		
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II						
SCALE	SHEET	UNITS	TYPE	DATE	DRAWN	
NTS	1 / 3	-	SCHEMATIC	29/09/2026		

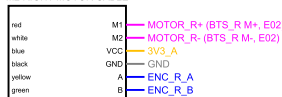


PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE				DWG No.	
POWER DISTRIBUTION				CR-V2-E02	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	TYPE	DATE	DRAWN
NTS	2 / 3	-	SCHEMATIC	29/09/2026	

M1 LEFT MOTOR CABLE



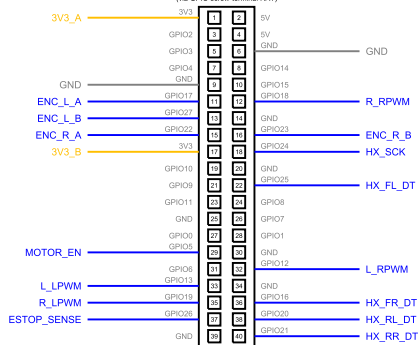
M2 RIGHT MOTOR CABLE



Encoder VCC = 3.3 V ONLY (5 V would put 5 V on the Pi's GPIO pins).

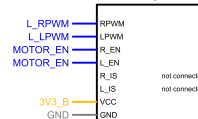
RASPBERRY PI 5 - 40-PIN HEADER

(via GPIO screw-terminal HAT)

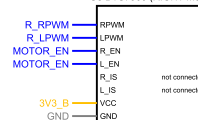


Do NOT use pins 2 / 4 (5 V) for anything: every signal here is 3.3 V. Unused GND pins may be used.

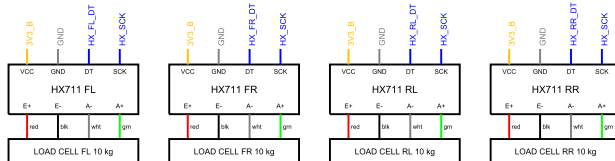
U2 BTS7960 (LEFT motor)



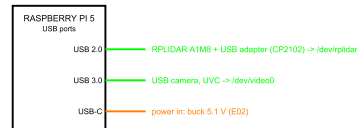
U3 BTS7960 (RIGHT motor)



BTS7960 VCC from 3.3 V makes its inputs 3.3 V-logic compatible. Power side (B+&B-M+): see E02.



FL/FR/RL/RR = front-left / front-right / rear-left / rear-right, looking down, front away from you.
A cell that reads negative: swap its A+/A- (green/white) or set its sign to -1 in robot.yaml.



ESTOP_SENSE comes from the R1/R2 divider on E02 (junction below R1).
Same net name = same wire. Wire IDs S1..S30: BUILD_GUIDE.md week 3.

PROJECT					
AUTONOMOUS CARGO ROBOT - DESIGN v2					
TITLE				DWG No.	
CONTROL & SENSOR WIRING				CR-V2-E03	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	TYPE	DATE	DRAWN
NTS	3 / 3	-	SCHEMATIC	29/09/2026	